

CTNT 2026, JUNE 5 - 7
SCHEDULE, TITLES, AND ABSTRACTS

TITLES AND ABSTRACTS (Alphabetical by last name)

Speaker: Patrick Allen (McGill University)

Title: Minimal lifts in the badly dihedral case

Abstract: An example of Serre shows that in the strong form of his modularity conjecture, one can't simultaneously demand minimal weight, level, and nebentypus. Serre and Carayol independently explained that the obstruction to a minimal nebentypus comes from points in the upper half plane with nontrivial stabilizers in $\mathrm{PSL}(2, \mathbb{Z})$, hence only occurs for the primes 2 and 3 and certain Galois representations that we'll call badly dihedral.

But for these badly dihedral representations, can one sometimes lift to minimal weight, level, and nebentypus? If so, can we understand when it happens? We'll answer these questions in the $p = 3$ case, connecting them to certain class groups via an $R = T$ theorem. This is joint work in progress with Preston Wake.

Speaker: Santiago Arango-Piñeros (UMass Amherst)

Title: A modern view of Fermat's method of descent

Abstract: In the 1650s, Pierre de Fermat introduced the method of *infinite descent* to prove, among other things, that *the sum of two biquadrates is never a biquadrate or a square*. In this talk, we will revisit Fermat's method of descent from the point of view of modern arithmetic geometry, highlight some more recent developments, and explain some interesting connections with the theory of modular curves.

Speaker: Sarah Arpin (Virginia Tech)

Title: Elliptic Curves and Isogeny-Based Cryptography

Abstract: This talk will give an introduction to isogeny-based cryptography through the study of isogeny graphs, built using maps between isomorphism classes of elliptic curves. We will survey several kinds of isogeny graphs, including graphs made from objects and maps defined over finite fields $\mathbb{F}_p$, over the algebraic closure $\overline{\mathbb{F}}_p$, and variants equipped with additional structure such as orientations and level structure. Along the way, we will see how these different perspectives lead to new cryptographic constructions.

Speaker: Mike Cerchia (University of Maine)

Title: Quadratic points on modular curves and images of Galois

Abstract: We discuss work with Rakvi on the classification of images of Galois associated to elliptic curves over quadratic fields, which is equivalent to the classification of all quadratic points on certain modular curves. The main result is the determination of modular curves of prime-power level that have infinitely many quadratic points. We also discuss progress on computing quadratic points on the resulting modular curves with finitely many such points.

Speaker: Huy Dang (Binghamton University)

Title: Lifting abelian covers from characteristic p to characteristic 0

Abstract: This is based on joint work with Khai-Hoan Nguyen-Dang. We discuss a bridge between Artin–Schreier–Witt theory in characteristic p and Kummer theory in mixed characteristic. The main construction uses Matsuda’s deformed Artin–Hasse exponential to recover the Sekiguchi–Suwa group schemes, which lift cyclic p -power torsors from characteristic p to characteristic 0. We explain how this construction relates to refined Swan conductors and degeneration of torsors.

Speaker: Juanita Duque-Rosero (Boston University)

Title: Stacky curves and local-to-global principles

Abstract: A classical question in number theory is the following: if an equation with integer coefficients has solutions modulo every prime and over the real numbers, must it also have an integer solution? When the answer is yes, this is an example of a local-to-global principle. In this talk, we will explore this question for generalized Fermat equations of the form $x^2 + By^2 = Cz^n$ for n odd, focusing on the existence of integral points. A surprising feature of this problem is that the natural geometric object attached to these equations is not an ordinary curve, but a stacky curve, which keeps track of extra symmetries at certain points. We will see how this perspective helps organize the arithmetic and leads to some unexpected results. The research is joint work with Christopher Keyes, Andrew Kobin, Manami Roy, Soumya Sankar, and Yidi Wang.

Speaker: Amanda Folsom (Amherst College)

Title: Quantum modular forms and applications

Abstract: The subject of quantum modular forms is relatively new, with its definition and initial examples introduced by Zagier in 2010. These functions are not modular in the traditional sense, but the ways in which they fail to be modular exhibit rich analytic behavior. In this talk, we will provide an accessible introduction to the subject of quantum modular forms, and some applications and problems of interest in number theory and other fields of mathematics.

Speaker: Tyler Genao (Ohio State University)

Title: Elliptic curves, torsion points and Galois representations

Abstract: Elliptic curves are special amongst algebraic curves: the set of points on an elliptic curve is an abelian group under a chord and tangent method, which gives a way to construct many new

points from just a few. Of particular interest is the torsion subgroup of an elliptic curve, which is the subgroup of points with finite order. Torsion points can generate interesting field extensions, in analogy to cyclotomic fields.

In this talk, we will learn about elliptic curve torsion points and how to study torsion via Galois representations, which help describe the rationality of these points. We will also see what type of research can be done on Galois representations to prove new results on elliptic curves, with a good amount of elementary group theory.

Speaker: Daniel Gotshall (University of Rochester)

Title: Hilbertian Fields and Profinite Iterated Monodromy Groups Over Transcendental Elements with Specializations of Infinite Index

Abstract: Given a field k of characteristic $p \geq 0$, a polynomial $f(x)/k$ of degree d with $p \nmid d$, and t transcendental over k , we can visualize the (necessarily disjoint) sets $f^{-n}(t)$ for $n \in \mathbb{N}$ as the vertices of a d -ary tree graph rooted at a . The permutations of the elements in these sets by $\text{Gal}(k(f^{-n}(t))/k(t))$ correspond to automorphisms of the tree graph. Similarly, one can replace t with $a \in k$ and consider $\text{Gal}(k(f^{-n}(a))/k)$ in an analogous way. Letting G_t denote the inverse limit of $\text{Gal}(k(f^{-n}(t))/k(t))$ and similarly for G_a , we might have $G_a \not\leq G_t$, depending on a ramification index. Nevertheless, we can abuse notation and define $[G_t : G_a] := |G_t|/|G_a| \in \mathbb{N} \cup \{\infty\}$. If k is Hilbertian, then setting $t = a$ in $g(x, t) = f(x) - t \forall a \in k \setminus A$ for some “thin” proper subset A preserves the irreducibility of $g(x, a)$. Hence one might think that $[G_t : G_a] < \infty$ in this case. To the contrary, every countable Hilbertian field k has a countable Hilbertian extension F such that $[G_t : G_a] = \infty \forall a \in F$ and any polynomial $f(x)/F$ such that G_t has infinitely many maximal normal subgroups. This is a generalization of a related result by Dittmann and Kadets.

Speaker: Catherine Hsu (Swarthmore College)

Title: Explicit $R = \mathbb{T}$ via rank bounds

Abstract: As illustrated by the Herbrand-Ribet theorem, the relationship between modular forms and Galois representations is a powerful tool for studying arithmetic properties of number fields. In this talk, I will discuss a project, joint with Preston Wake and Carl Wang-Erickson, that explores the connection between a Galois deformation problem arising from Eisenstein congruences and the arithmetic of certain number fields with twisted-Heisenberg Galois group. We relate the rank of a certain Hecke algebra to an arithmetic invariant arising from the vanishing of a triple Massey product and deduce new $R = \mathbb{T}$ results in some cases in which Wiles’ numerical criterion fails.

Speaker: Eray Karabiyik (Cornell University)

Title: Classifying Hyperelliptic Modular Curves

Abstract: Let G be an open subgroup of $\text{GL}_2(\widehat{\mathbb{Z}})$ satisfying $\det(G) = \widehat{\mathbb{Z}}^\times$ and $-I \in G$. Associated to G , there is a modular curve X_G defined over $\mathbb{Q}$ which parametrizes elliptic curves with G -level structure. In this talk, we characterize all geometrically hyperelliptic modular curves and classify those that are hyperelliptic. We also classify all modular curves of genus 0 up to $\mathbb{Q}$ isomorphism.

Speaker: **Kiran S. Kedlaya** (University of California San Diego)

Title: A formalizable approach to classifying vector bundles on the Fargues-Fontaine curve

Abstract: For any local field E , geometrization of the local Langlands correspondence involves the study of vector bundles on a certain “curve” (= regular one-dimensional scheme) associated to E , and in particular the classification of indecomposable bundles. When $E = \mathbb{C}$ this is the usual projective line over E , and it was shown by Grothendieck that the indecomposable bundles are precisely the line bundles. When $E = \mathbb{R}$, the relevant curve is the pointless conic $x^2 + y^2 + z^2 = 0$ considered by Simpson. For other local fields the relevant curve was introduced by Fargues-Fontaine, but most proofs of the classification of indecomposables rely on one of several deep p -adic ingredients. We describe a proof using only easily stated scheme-theoretic properties of the curve, which should be feasible for formalization in Lean (although we will not discuss formalization in the talk itself). Based on joint work with Peter Dillery.

Speaker: **Jacob Mayle** (University of Delaware)

Title: Coprime Reductions of Elliptic Curves

Abstract: A classical question in number theory asks how often two “random” integers are relatively prime. In this talk, we discuss an elliptic curve analogue: for two non-CM elliptic curves over $\mathbb{Q}$, how often do their reductions modulo p have coprime order? We formulate a conjecture for the density of such primes, motivated by Zywin’s refinement of the Koblitz conjecture, and present numerical evidence and average results obtained by studying Serre pairs. This is joint work with Asimina S. Hamakiotes, Sung Min Lee, and Tian Wang.

Speaker: **Sung Min Lee** (Wake Forest university)

Title: How Often Are Elliptic Curve Reductions Divisible by m ?

Abstract: Let $E/\mathbb{Q}$ be an elliptic curve, and let E_p denote its reduction modulo a prime p of good reduction. A natural question is how the group orders $\#E_p(\mathbb{F}_p)$ are distributed as p varies. In particular, for a fixed integer m , one may ask how often m divides $\#E_p(\mathbb{F}_p)$. A naive random-integer heuristic suggests that the density should be $1/m$.

In this talk, I will discuss joint work with Nara Sheen on the density of m -divisibility primes. For Serre curves, we compute this density explicitly and show that it is greater than $1/m$ for every integer $m \geq 2$. We also compute the corresponding average constant over families of elliptic curves ordered by height, with explicit error bounds. These results show that, for a generic elliptic curve, the sequence of group orders $\#E_p(\mathbb{F}_p)$ is divisibility-biased: it is more likely to be divisible by m than a random integer is.

Speaker: **Daniel Litt** (University of Toronto)

Title: Arithmetic of differential equations

Abstract: I'll survey some conjectures on relationships between the arithmetic of differential equations and the algebraicity of their solutions—notably the Grothendieck-Katz p -curvature conjecture, and discuss some recent developments on these topics. The talk will include joint work with Josh Lam.

Speaker: James Austin Myer (CUNY Graduate Center)

Title: How to Construct a Regular Model of a Hyperelliptic Curve in Mixed Characteristic $(0, 2)$

Abstract: We learn early on the power of considering an elliptic curve modulo a prime. This technique is particularly amenable for primes of good reduction. However, there are always some bad primes: those which divide the discriminant. Upon clearing denominators, we may regard the equation of an elliptic curve as describing a *family* of elliptic curves: one over each prime, & thus, a surface. While we cannot reasonably ask that every prime is of good reduction, we can ask instead that the surface worth of elliptic curves is analogous to a manifold, i.e. regular. Often, the surface immediately obtained by clearing denominators is not regular. So, we have to perform some surgery to make it regular, i.e. blow up the surface. Any regular such surface is called a *regular model* of the curve. The local problem is simpler than the global, & so we localize, & consider the problem one-prime-at-a-time, i.e. in *mixed characteristic*. A hyperelliptic curve is a natural generalization of an elliptic curve. We discuss an algorithm to construct a regular model of a hyperelliptic curve in mixed characteristic $(0, 2)$. The novelty lies in completing the story for the only odd prime, which is even more odd than we would have thought.

Speaker: Suzanne O'Hara (Wesleyan University)

Title: Heavenly Elliptic Curves over Cubic Fields

Abstract: We call an elliptic curve E/K heavenly at ℓ when the field $K(E[\ell^\infty])$ is a pro- ℓ and unramified away from ℓ extension of $K(\mu_\ell^\infty)$. This puts extra arithmetic constraints on E that guarantees an upper-triangular representation of G_K on the ℓ -torsion of the curve. Previous work has shown that in many cases, for a fixed field K there are only finitely many K -isomorphism classes of heavenly elliptic curves when we let ℓ vary through all primes. Recent work of McLeman and Rasmussen explores a “bi-directional” finiteness result where both the prime of interest and the quadratic field of definition are allowed to vary. In this talk we will discuss analogues of their results for elliptic curves defined over cubic number fields. We will also discuss under what conditions the trace of a heavenly elliptic curve behaves like the trace of a curve with complex multiplication.

Speaker: Jen Paulhus (Mount Holyoke College)

Title: Completely decomposable Jacobian varieties

Abstract: Thirty years ago, Ekedahl and Serre posed two intriguing questions about the decomposition of Jacobian varieties. They asked if, for every genus, there is a curve whose Jacobian variety decomposes into only elliptic curves and, barring that, if there is an upper bound to the genera for which such a curve exists. Using several different techniques they gave examples up to genus 1297

(but still with many unknown genera less than 1297). I will talk about work that has been done since then, mainly computational, to find many more examples.

Speaker: Rakvi (The University of Maine)

Title: On congruence obstructions in the refined Koblitz conjecture

Abstract: Let $E/\mathbb{Q}$ be an elliptic curve. Zywin's refinement of Koblitz's conjecture predicts that there are infinitely many primes p of good reduction for which $\#E_p(\mathbb{F}_p)$ is prime precisely when there is no congruence obstruction to this occurring. In this talk, I will discuss an ongoing joint work with Sung Min Lee and Jacob Mayle in which we aim to prove two results motivated by this conjecture. First, assuming GRH, we give a lower bound on the least prime divisor of $\#E_p(\mathbb{F}_p)$ for infinitely many primes p of good reduction. Second, we give a complete characterization of elliptic curves over $\mathbb{Q}$ with a primitive congruence obstruction modulo a product of two distinct primes.

Speaker: Ravi Ramakrishna (Cornell University)

Title: Applications of cutting asymptotically good towers of number fields

Abstract: Given an infinite profinite extension K_∞ of a number field K , one can ask if it is asymptotically good, that is for any L in K_∞ , is the root discriminant of L/K uniformly bounded? I'll explain the current records for $K = \mathbb{Q}$ in the pro- p setting and how cutting was used to get them, to answer an old question of Ihara, and its role in the recent new lower bound for the Erdős unit distance problem. This is joint work with F. Hajir and C. Maire.

Speaker: Joseph H. Silverman (Brown University)

Title: Number Theory, Dynamical Systems, and Wide Spacing of Orbits

Abstract: An abstract discrete dynamical system consists of a set X and a self-map $f : X \rightarrow X$. Dynamics studies properties of orbits of points of X under repeated application of f . When X and f are defined by polynomials, algebraic geometry comes into play, and when the polynomials have integer coefficients, number theory joins the game. In the first part of this talk, I will discuss how these three areas of mathematics, dynamical systems, algebraic geometry, and number theory, have come together in the past 30+ years to form the active new field of **arithmetic dynamics**. In the second part I will give a taste of arithmetic dynamics through a recent conjecture which says that in the number theory setting, there are lots of orbits of rational points and that they are widely spaced throughout the space X .

Speaker: Caylee Spivey (University of Connecticut)

Title: Counting k -free ideals in generalized arithmetic progressions

Abstract: A classical problem in number theory is to estimate the number of integers $1 \leq n \leq X$ that are not divisible by a k -th power and that lie in a fixed arithmetic progression, that is $a \equiv b \pmod{m}$. In arbitrary number fields, there have been estimates made on the number of k -free ideals

with norm up to X . In this talk, we will combine these two problems: we seek an estimate on the number of k -free ideals of bounded norm in generalized arithmetic progressions in number fields. This is ongoing work that began at the sixth Rethinking Number Theory workshop.

Speaker: Connor Stewart (CUNY Graduate Center)

Title: Conductor-Discriminant Inequality for Tame Ramified Cyclic Covers

Abstract: We consider $\mathbb{Z}/n$ -covers $X \rightarrow \mathbb{P}^1$ defined over discretely valued fields K with excellent valuation ring $\mathcal{O}_K$ and perfect residue field of characteristic not dividing n . Two standard measures of bad reduction for such curves X are the Artin conductor of the minimal regular model of X over $\mathcal{O}_K$ and the valuation of the discriminant of an integral Weierstrass equation for X . We prove an inequality relating these two measures. Specifically, if X is given by an affine equation $y^n = f(x)$ with $f(x) \in \mathcal{O}_K[x]$, and if $\mathcal{X}$ is its minimal regular model over $\mathcal{O}_K$, then the negative of the Artin conductor of $\mathcal{X}$ is bounded above by $(n-1)v_K(\text{disc}(\text{rad}(f)))$. This extends previous work of Ogg, Saito, Liu, Srinivasan, and Obus–Srinivasan on elliptic and hyperelliptic curves. (Joint work with Andrew Obus and Padmavathi Srinivasan.)

Speaker: Benjamin York (Skidmore College)

Title: The Adelic Galois Image of an Elliptic Curve with CM

Abstract: Let E be an elliptic curve defined over $\mathbb{Q}$. The Galois action on the torsion points of E defines a Galois representation $\rho_E: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}(2, \widehat{\mathbb{Z}})$, called the adelic Galois representation of E . Understanding the image of this representation (up to conjugation) is at the heart of Mazur’s so called “Program B.” In 2022, Zywina published an algorithm to compute the image of ρ_E for any elliptic curve $E/\mathbb{Q}$ without complex multiplication (CM). In this talk, we outline an algorithm to compute the image of ρ_E for an elliptic curve $E/\mathbb{Q}$ with CM whose j -invariant is not 0 or 1728. We will do so by studying entanglements between division fields of CM elliptic curves and by studying how twists affect these entanglements. This work is joint with Álvaro Lozano-Robledo.

CTNT 2026 -- June 5-7 -- Titles		
FRIDAY, JUNE 5TH		
Time	Event in room BPB 131	Titles for Plenary Talks
12:45-1:00	<i>Welcoming remarks</i>	
1-1:45	Tyler Genao	Elliptic curves, torsion points and Galois representations
2:00-2:45	Sara Arpin	Elliptic Curves and Isogeny-Based Cryptography
3:00-3:45	Joseph Silverman	Number Theory, Dynamical Systems, and Wide Spacing of Orbits
4:00-4:30	<i>Break</i>	
	Event in room BPB 131	Event in room BPB 130
4:30-4:50	Mike Cerchia	Jacob Mayle
5:00-5:20	Caylee Spivey	Daniel Gotshall
5:30-6:00	Eray Karabiyik	Sung Min Lee
SATURDAY, JUNE 6TH		
Time	Event in room BPB 131	Titles for Plenary Talks
8:30-9:00	<i>Refreshments/Coffee</i>	
9:00-9:45	Amanda Folsom	Quantum modular forms and applications
10:00-10:45	Patrick Allen	Minimal lifts in the badly dihedral case
11:00-11:30	<i>Break</i>	
11:30-12:15	Cathy Hsu	Explicit $R=T$ via rank bounds
12:30-2:00	<i>Lunch break</i>	
2:00-2:45	Daniel Litt	Arithmetic of differential equations
3:00-3:45	Juanita Duque-Rosero	Stacky curves and local-to-global principles
4:00-4:30	<i>Coffee Break</i>	
	Event in room BPB 131	Event in room BPB 130
4:30-4:50	Benjamin York	Suzanne O'Hara
5:00-5:20	Rakvi	Huy Dang
5:30-6:00	Connor Stewart	James Myer
SUNDAY, JUNE 7TH		
Time	Event in room BPB 131	Titles for Plenary Talks
8:30-9:00	<i>Refreshments/Coffee</i>	
9:00-9:45	Ravi Ramakrishna	Applications of cutting asymptotically good towers of number fields
10:00-10:45	Santiago Arango Pineros	A modern view of Fermat's method of descent
11:00-11:30	<i>Break</i>	
11:30-12:15	Jen Paulhus	Completely decomposable Jacobian varieties
12:30-1:15	Kiran Kedlaya	A formalizable approach to classifying vector bundles on the Fargues-Fontaine curve
1:15-1:30	<i>Concluding Remarks</i>	